

Exam Date & Time: 03-Jan-2025 (02:30 PM - 05:30 PM)



# MANIPAL ACADEMY OF HIGHER EDUCATION

**Manipal Institute of Technology, Bengaluru**  
BTech VII Semester - Make up Examination, Jan-2025

## Digital Forensics [IT-3154]

**Marks: 50**

**Duration: 180 mins.**

### PART-A

**Answer all the questions.**

- 1) During a cyber incident investigation, a forensic analyst must acquire data from a suspect's laptop without altering any original files. Explain data acquisition techniques and tools should the analyst use to preserve data integrity, and identify why these methods are essential in a legal investigation? (5)
  - A)
  - B)
- 2) A company suspects that sensitive information has been leaked through unauthorized access to its internal network. As a forensic investigator, you are assigned to trace the source of the breach and gather evidence of data exfiltration. Choose network forensic techniques would you use to identify the source and track data flow? Illustrate would you differentiate between legitimate and suspicious network traffic to support your investigation? (5)
  - A)
  - B)
- 3) You arrive at a crime scene involving suspected financial fraud, where multiple electronic devices, including a laptop and several external drives, are found. Explain what steps would you take to secure and document the scene, and how would you handle and collect the devices to preserve digital evidence integrity? (5)
  - A)
  - B)
- 4) During a digital forensic investigation, an intercepted message "UDKXCQ" is suspected to be encrypted using a Vigenère cipher with the repeating key "SECURE". Investigators know the message could contain either meaningful text or random characters. Explain (5)

and decrypt the message to reveal the original plaintext

- A)
  - B) A company is investigating an employee suspected of leaking confidential information via their mobile device and work email account. The employee's phone and email records are accessible for analysis. Illustrate the steps you would take to examine the mobile device and email logs to uncover any evidence of data leakage, focusing on methods to recover deleted messages and trace email communication. (5)
- 5) A cybersecurity team is tasked with assessing the potential vulnerabilities of a company's public-facing web server. Your objective is to gather as much information as possible about the server and its infrastructure without directly interacting with it.
- A) Compare and contrast the footprinting and reconnaissance techniques you would use to gather details on the server's IP address, DNS records, and possible open ports, while remaining undetected. (5)
  - B) A company detects unusual traffic on its network and suspects that a malware infection is causing data leakage. The malware may be using sniffing techniques to capture sensitive information from unencrypted communications. Explain how would you identify the presence of sniffing malware on the network, and what steps would you take to contain and remove it? (5)

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